Net_Practice

My solution and basic explanation on Net_Practice and networking basics.

First of all, thanks to Robin | radelwar for explaining the more in depth stuff to me.
This may not be perfect since I just learned about all of the following myself. So please, if there is anything wrong or missing, please notify me by reaching out to me (my profile) or create an issue, thanks.

I only created this tutorial, since the project has a few flaws and you have the ability to solve it in a way that's wrong, but the project is still telling you that everything is correct.
And all other tutorials I found used these flaws to simplify the solution.

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Basics

For this project we only use IPv4, so I won't talk about IPv6.
An IPv4-adress is a 32-bit number divided into 4 "blocks", each 8 bits.
i.e.:
`192.168.100.1` turns into `11000000.10101000.01100100.00000001`
So the min. value of one "block" is `0` and the max. value is `255`.
The same logic applies to the network-mask:
`255.255.255.0` turns into `11111111.11111111.11111111.00000000`
Special to the mask is, after one bit was `0` there can't be any `1` bit's anymore.
So the only available numbers are:

- `255 (binary: 11111111)`
- `254 (binary: 11111110)`
- `252 (binary: 11111100)`
- `248 (binary: 11111000)`
- `240 (binary: 11110000)`
- `224 (binary: 11100000)`
- `192 (binary: 11000000)`

- `128 (binary: 10000000)`
- `0 (binary: 00000000)`

Through which `255.255.255.0` is a valid mask
and `255.255.128.128` is **not** a valid mask.

In order to have the ability to send packages between two IP-addresses they either need to be part of the same network or they need to be connected by a router which is part of both subnets.

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Special IP-ranges

The following special address-ranges are reserved for Private Networks:
`10.0.0.0 – 10.255.255.255`
`172.16.0.0 – 172.31.255.255`
`192.168.0.0 – 192.168.255.255`

The following address-range is reserved for so called loopback addresses:
`127.0.0.0 – 127.255.255.255`

There is some more special ip-ranges, but for this project, you only need to remember those above.

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Masks

The network-mask, subnet-mask or in our project only called mask is there to decide which range of ip-adresses are part of the same subnet.
There are 2 different ways of writing the mask:

- "Dot-decimal notation": `255.255.255.0`
- "Class Inter-Domain Routing" or "CIDR": `/24`

The more usable ip-addresses you need in one subnet, the less subnets you will be able to create.
To help you understanding it, I found this table very helpful:

CIDR	Dot-decimal	Number of IP-addresses per subnet	Usable IP-addresses per subnet	Number of subnets
/32	255.255.255.255	1	0	256
/31	255.255.255.254	2	0	128
/30	255.255.255.252	4	2	64
/29	255.255.255.248	8	6	32
/28	255.255.255.240	16	14	16
/27	255.255.255.224	32	30	8
/26	255.255.255.192	64	62	4
/25	255.255.255.128	128	126	2
/24	255.255.255.0	256	254	1

The number of usable IP-addresses per subnet is lower than the total number of IP's because the first address is reserved as the network-address of the subnet and the last address is reserved as a broadcast-adress.
i.e. for mask `255.255.255.252`:
network: `190.3.2.252`
broadcast: `190.3.2.255`
usable IP's: `190.3.2.253`, `190.3.2.254`

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Switches

A switch will enable you to connect more than two devices to the same network.
Its only purpose is to distribute packages to its network.
To see a working example, you can take a look at Level 3.

Routers

As previously mentioned a router is an interface which enables communication between different networks.
A router has the ability to be part of multiple networks, in Netpractice this is visualized by the so called `Interface`.
If routers and switches are still magic to you, I suggest looking deeper into it yourself, as their basic understanding is crucial to succeed in this project.

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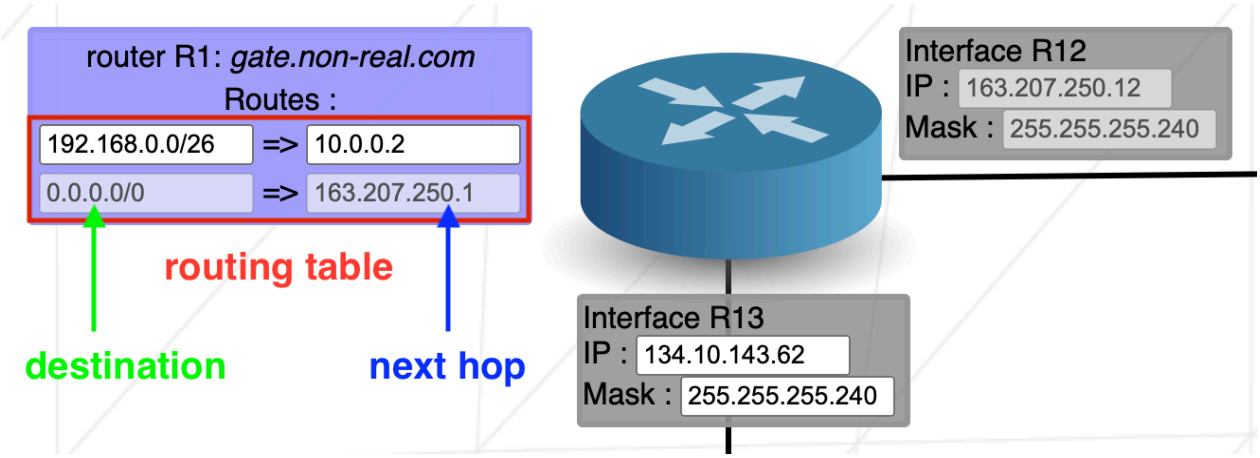
Routing Table



The routing table is there to store all the different paths to all the networks, the device is part of.
In Net_Practice the routing table consists of two elements, the **destination** and the **next hop**
The **destination** consists of the network-address that you want to send a package to, combined with the CIDR of that network: `190.3.2.252/30`. If you don't want to specify a destination, you can just set it to `default` or `0.0.0.0/0`.
The **next hop** is the address of the next router that you need to send the packages to in order to reach the destination-network.

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Network

And now to connect all of the above mentioned topics.
In order to have a functioning network, you now need to apply all of the parts talked about earlier.
If there should be a working connection in a network, the devices somehow need to be connected, either directly or with the help of routers which are part of both networks.

Now you may ask, how do I know if two devices are part of the same network?
For this you need to combine the IP-address and the mask of the devices in order to get the network-adress, that device is part of.
By combining I mean, doing a bit-by-bit-AND-opperation.
For that we first need to translate the IP and the mask to binary.
i.e.:
IP: `192.168.100.1` in binary: `11000000.10101000.1100100.00000001`
MASK: `255.255.255.0` in binary: `11111111.11111111.11111111.00000000`
Now you just combine the two bit by bit, if both bits are a `1` the corresponding bit of the network-address is `1`, in any other case the corresponding bit is `0`.

By doing that to the mentioned example, you should get the network-address of
`11000000.10101000.1100100.00000000` in binary or `192.168.100.0` in dot-decimal.
If two devices share the same network-address, they are part of the same network and communication is ensured.

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Levels

Here are all the solutions and explanations for all 10 Levels.

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